Anaerobe 46 (2017) 41–45

Contents lists available at ScienceDirect

Anaerobe

journal homepage: www.elsevier.com/locate/anaerobe

Anaerobes in the microbiome

# Simple biogas desulfurization by microaeration – Full scale experience

P. Jeníček [a], J. Horejš [b], L. Pokorná-Krayzelová [a, *], J. Bindzar [a], J. Bartáček [a]

[a] Department of Water Technology and Environmental Engineering, University of Chemistry and Technology Prague, Technická 5, 166 28, Prague 6, Czechia
[b] K&K Klatovy, Zlatnická 33, 339 01, Klatovy, Czechia

ARTICLE INFO

*Article history:*
Received 23 October 2016
Received in revised form
25 December 2016
Accepted 4 January 2017
Available online 5 January 2017

Handling Editor: Kornel L. Kovacs

*Keywords:*
Anaerobic digestion
Biogas
Hydrogen sulfide removal
Microaeration
Sludge stabilization

ABSTRACT

Hydrogen sulfide in biogas is common problem during anaerobic treatment of wastewater with high sulfate concentration (breweries, distilleries, etc.) and needs to be removed before biogas utilization. Physico-chemical desulfurization methods are energetically demanding and expensive compare to biochemical methods. Microaeration, i.e. dosing of small amount of air, is suitable and cost effective biochemical method of sulfide oxidation to elemental sulfur. It has been widely used in biogas plants, but its application in anaerobic reactors for wastewater treatment has been rarely studied or tested. The lack of full-scale experience with microaeration in wastewater treatment plants has been overcome by evaluating the results of seven microaerobic digesters in central Europe. The desulfurization efficiency has been more than 90% in most of the cases. Moreover, microaeration improved the degradability of COD and volatile suspended solids.

© 2017 Elsevier Ltd. All rights reserved.

## 1. Introduction

Anaerobic digestion of sludge is the key process for municipal wastewater treatment plants aiming to exploit the energy potential of organic pollution of wastewater through biogas [10,20]. The suitable quality of biogas is the limiting condition for effective power production and also for all other types of biogas utilization. The elevated concentration of hydrogen sulfide is the most frequent problem.

Many physico-chemical methods of biogas desulfurization are available, such as alkali washout, sorption, precipitation, stripping, etc. Operation at high temperature and pressure, as well as the need for additional equipment and chemicals, make physico-chemical methods energetically demanding and expensive [1,16]. In contrast, biological methods based on the biochemical oxidation of sulfide to sulfate, thiosulfate and elemental sulfur involve lower operational costs with lower or no need for chemical addition [2,21]. Among biological methods, microaeration is the simplest one. It can be performed inside the anaerobic reactor without requirements to build a new separate desulfurization unit.

Microaeration has recently proved to be highly efficient biological method of sulfide removal regarding anaerobic digestion of wastewater and sludge [4,9,12,15,22,24]. This process is based on the controlled dosing of a limited amount of air or oxygen into the digester to ensure the oxidation of sulfide into elemental sulfur. Precipitated elemental sulfur is removed from the digester together with the digested sludge. The presence of oxygen does not negatively influence the activity of anaerobic bacteria because it is consumed quickly and almost completely [6].

A variable air/oxygen dosing rate is necessary as the consequence of the feed composition and loading rate variations resulting in the varying production of sulfide. Besides, residual oxygen in the biogas must meet the requirements of the biogas utilization technology that will be employed afterwards. Optimal process control is the key parameter to the successful microaeration in such cases. When the digestion process is stable as regards biogas production and $H_2S$ concentration, the air/oxygen flow rate can be constant for certain period. However, when it is necessary to cope with the changes of $H_2S$ concentration and biogas flow, oxygen supply must be controlled [14].

Until now, the lack of full-scale experience in wastewater treatment plant digesters was reported as a disadvantage of the microaerobic desulfurization [14]. Present work describes the experiences obtained during long-term operation (in the period of

* Corresponding author.
*E-mail address:* krayzell@vscht.cz (L. Pokorná-Krayzelová).

http://dx.doi.org/10.1016/j.anaerobe.2017.01.002
1075-9964/© 2017 Elsevier Ltd. All rights reserved.

years 2003–2015) of 7 microaerobic digesters in central Europe in terms of $H_2S$ removal and changes in the quality of sludge and sludge liquor.

## 2. Material and methods

Microaeration has been applied on seven municipal wastewater treatment plants (WWTPs) in central Europe. Their characterization is described in Table 1. In each of them a desulfurization unit (Fig. 1) has been built to blow air inside of the digester. The desulfurization unit consisted of the source of pressurized air (such as air compressor), and the control unit which was able to monitor and control the air flow. This simple technology does not need changes in the digester construction and can be implemented without interruption of the digestion process. In all evaluated digesters, air has been dosed into the recirculation stream (Fig. 2).

## 3. Results and discussion

### 3.1. Start-up

Start-up period of microaerobic digestion took usually 3–12 weeks as shown in Table 2. Nevertheless, our experiences from laboratory and pilot experiments indicated, that the start-up period can be much shorter if necessary (data not shown). Following aims must be fulfilled during start-up period: (1) the optimization of air dose, and (2) the adaptation and growth of sulfide oxidizing bacteria. Fig. 3 shows the start-up of digester U. Microaeration was turned on in the week 29 and the start-up took approx. 12 weeks.

As expected, the dose of air depends strongly on $H_2S$ concentrations in both gaseous and liquid phase. In evaluated digesters the air doses were in the range of hundreds and thousands of litres per hour in general, no cleaning or pretreatment of air was necessary.

At higher air doses nitrogen from air will dilute biogas and decrease the methane concentration. In microaerobic digesters monitored in present work the volumetric dose of air ranged between 1 and 3% of biogas production. In such cases the effect of the remaining $N_2$ on the biogas composition was negligible – maximum decrease of methane concentration did not exceed 2%. Therefore it was not necessary to consider replacement of air by expensive pure oxygen [4].

### 3.2. Desulfurization efficiency

The achieved desulfurization efficiency was between 74% and 99% (Table 2). Requested reduction of $H_2S$ concentration was reached in all cases. In addition long-term operation brought usually further decrease of the $H_2S$ concentration.

The typical course of $H_2S$ concentration in biogas when microaeration technology is applied (for digester K) illustrates Fig. 4. The desulfurization efficiency is effective even with strong fluctuation of initial sulfide and/or $H_2S$ concentration which is a common problem at WWTPs. From operational point of view high efficiency and stability of desulfurization is important. Hydrogen sulfide compromises the functions of cogenerations unit, causes the corrosion of concrete and steel, and is toxic to humans.

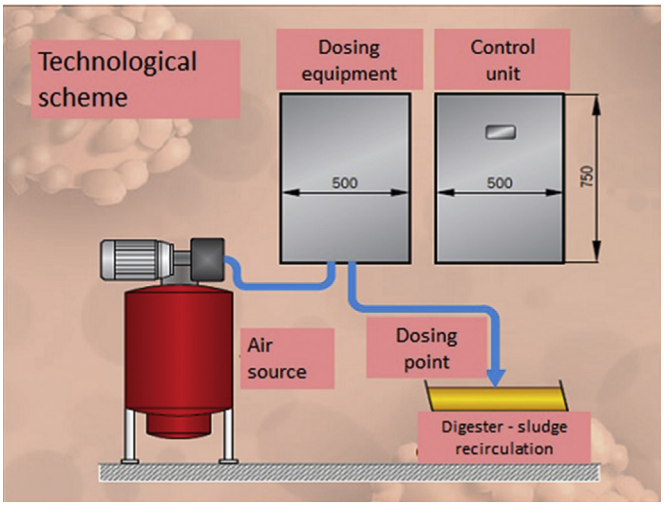



**Fig. 1.** Technological scheme of full-scale desulfurization unit (designed by K&K Klatovy).

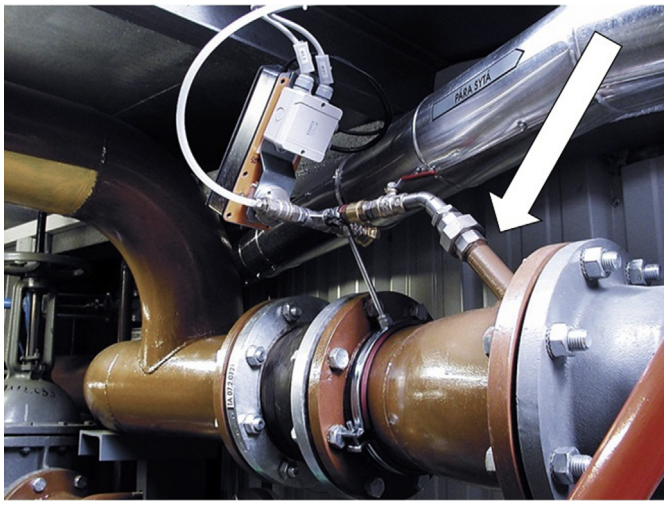


**Fig. 2.** Air dosing point into the sludge recirculation pipe.

### 3.3. Quality change of digested sludge and sludge liquor

It was indicated that microaeration is able to improve not only the biogas quality but the quality of digested sludge and sludge liquor as well [7]. The most important changes were found in the

**Table 1**
Characterization of selected municipal WWTP.

| Digester | Population equivalent [–] | Anaerobic reactor volume [m3] | Type of sludge [–] | The start of microaeration [year] |
|---|---|---|---|---|
| P | 19 000 | 1600 | primary + waste activated | 2012 |
| M | 29 000 | 2600 | primary + waste activated | 2012 |
| K | 19 000 | 1900 | primary + waste activated | 2010 |
| Ko | 35 000 | 3200 | primary + waste activated | 2010 |
| L | 27 000 | 3000 | primary + waste activated | 2005 |
| B | 350 000 | 30 000 | primary + waste activated | 2005 |
| U | 36 000 | 2100 | primary + waste activated | 2003 |

**Table 2**
Start-up results of microaerobic desulfurization in selected WWTP.

| Digester | Biogas production [$m^3d^{-1}$] | Air dose [$m^3h^{-1}$] | Time of start-up[a] [weeks] | $H_2S$ without microaeration[b] [mg $m^{-3}$] | $H_2S$ with microaeration[b] [mg $m^{-3}$] | $H_2S$ removal efficiency [%] |
|---|---|---|---|---|---|---|
| P | 750 | 1.00 | 7 | 3710 | 449 | 87.9 |
| M | 950 | 0.34 | 8 | 771 | 31 | 96.0 |
| K | 520 | 0.28 | 4 | 890 | 48 | 94.6 |
| Ko | 1000 | 1.00 | 3 | 3823 | 198 | 94.8 |
| L | 900 | 1.20 | 10 | 4383 | 21 | 99.5 |
| B | 9600 | 6.00 | 8 | 2633 | 690 | 73.8 |
| U | 830 | 1.20 | 12 | 7580 | 72 | 99.1 |

[a] Time of the air dose optimization.
[b] Annual average.

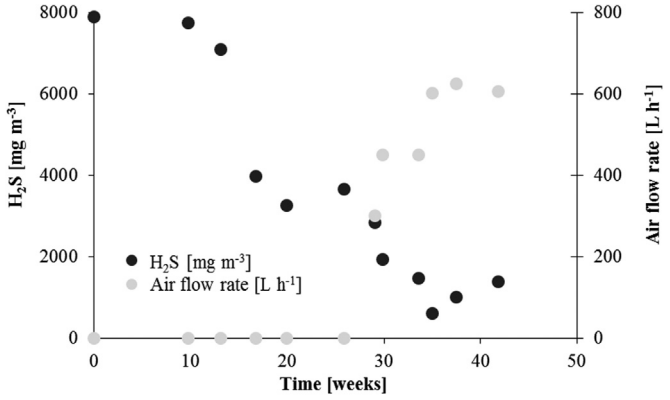



**Fig. 3.** Relation between hydrogen sulfide concentration of biogas and air flow rate (digester U).

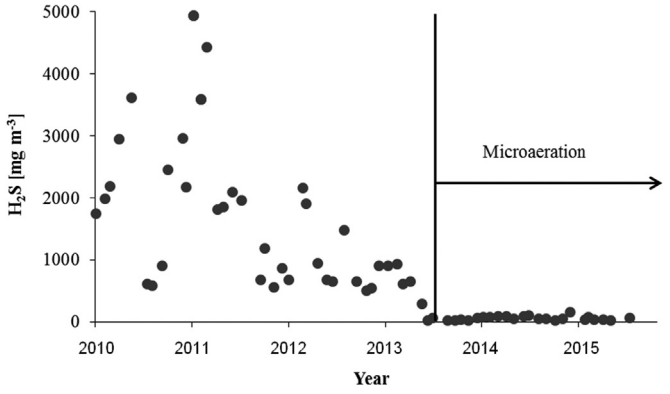



**Fig. 4.** $H_2S$ concentration in biogas before and after microaeration technology is applied (digester K).

soluble COD concentration of sludge liquor. The substantial significant decrease of the soluble COD was probably observed because of the combination of anaerobic and aerobic conditions which allows combining the degradation activity of anaerobic and aerobic microorganisms. The decrease of soluble COD was observed in all cases (Table 3) and varied between 18 and 33%. The combination of aerobic and anaerobic conditions can improve the hydrolysis; however, Ref. [11] observed that the extra hydrolysed products were oxidized to carbon dioxide.

Concerning $N_{ammon}$ concentration in the sludge liquor the results are rather ambiguous. It was observed both, either slight decrease in concentration or statistically unchanged concentration (Table 4). The influence of microaeration is not very strong in this case and it is combined with other factors such as pH changes,

**Table 3**
Change of average soluble COD concentration in sludge liquor due to microaeration.

| Digester | Soluble COD (average and standard deviation) | | |
|---|---|---|---|
| | Without microaeration [mg $L^{-1}$] | With microaeration [mg $L^{-1}$] | Decrease [%] |
| K | 373 ± 27 | 307 ± 32 | 17.7 |
| L | 430 ± 37 | 337 ± 36 | 21.6 |
| M | 340 ± 24 | 232 ± 20 | 31.8 |
| U | 778 ± 43 | 522 ± 39 | 32.9 |

increased $N_{ammon}$ release if degradation of organic matter is deeper, and the different specific gas loading rate of digester surface. Relatively low $N_{ammon}$ concentration caused by the low extent of sludge thickening before digestion could also affect the results. Ref. [3] reported significant decrease of $N_{ammon}$ concentration (about 10%) at values between 2000 and 2700 mg/L.

Another important finding is better volatile suspended solids (VSS) degradation during microaeration resulting in the decreasing ratio of volatile suspended solids to total suspended solids (VSS/TSS) in digested sludge (Table 5). Although this parameter is very good and sensitive indicator of digestion process, due to seasonal changes of the raw sludge quality, however, the fluctuation of this parameter occurs in the range of 2–5% in monitored digesters. The decrease is then strictly speaking statistically not significant. Given this context, we consider these changes in annual averages VSS/TSS as remarkable. Deeper degradation of sludge can be related to a potential inhibition of methanogens by sulfide. The oxidation of sulfide should eliminate such inhibition. This hypothesis could be supported by the fact that the highest decrease of VSS/TSS ratio was achieved at WWTP with the highest $H_2S$ concentration (digester U).

Even at the lowest decrease of VSS/TSS ratio presented in digester M it was possible to find the significant increase of biogas production. Because of the substantial yearly fluctuation of sludge quality and biogas production the respective monthly average production is compared in Fig. 5. The average increase of biogas production was 17%. On the top of it the average increase of methane production was 16%.

**Table 4**
Change of average $N_{ammon}$ concentration in sludge liquor due to microaeration.

| Digester | $N_{ammon}$ | | |
|---|---|---|---|
| | Without microaeration [mg $L^{-1}$] | With microaeration [mg $L^{-1}$] | Change [%] |
| K | 685 ± 28 | 690 ± 35 | +0.7 |
| L | 740 ± 57 | 718 ± 42 | −3.0 |
| M | 521 ± 34 | 487 ± 41 | −6.5 |
| U | 856 ± 32 | 784 ± 26 | −8.4 |

**Table 5**
Change of average VSS/TSS ratio of digested sludge.

| Digester | VSS/TSS | | |
|---|---|---|---|
| | Without microaeration [mg $L^{-1}$] | With microaeration [mg $L^{-1}$] | Decrease [%] |
| K | 0.629 | 0.612 | 2.7 |
| L | 0.566 | 0.546 | 3.5 |
| M | 0.515 | 0.507 | 1.6 |
| U | 0.658 | 0.597 | 9.3 |

### 3.4. Operational remarks

According to some authors, microaeration takes place almost solely in reactor headspace [5,13,18,19]. The yellow-whitish deposition of elemental sulfur on the walls and pipes can clog the system resulting in headspace overpressure and biogas leakage. In the digesters monitored and described in present work no problems with clogging of the gas pipes were observed [8]. The reason is probably strict dosing into the liquid phase where most of the oxygen is consumed and sufficient digester headspace where the rest of oxygen is consumed.

The air dosing point is often discussed topic of technological aspects of microaeration [14]. In present paper, all digesters applied microaeration with air dosed in the recirculation stream. In such case a bigger amount of air is required as a certain part of oxygen is consumed by other oxidation processes in addition to $H_2S$ removal. While applying the air into the gas phase of the digester, the advantage is the lower amount of air required, because of the lower extent of dissolved sulfide oxidation. On the other hand, problems with the precipitation of the elemental sulfur in gas pipes and other improper places may occur frequently [17].

In general, oxygen or air is not allowed to mix with the digester biogas because of the explosion risks of methane/oxygen mixture. However, the inflammable range is typically 5–15% by volume of methane with air [1,23]. Therefore, the leakage of biogas in air should be considered as the higher threat compare to the mixing of a small amount of air/oxygen with biogas. During common microaerobic desulfurization the amount of oxygen in biogas can never reach these values. In the evaluated digesters, the $O_2$ maximum concentration in biogas was 0.3%, however the amount was often below detection limit.

Lastly, it is necessary to say that present work does not intend to argue the application of the microaerobic process every time in each anaerobic process. Only well substantiated benefit and improvement of process such as the desulfurization of biogas, the suppression of sulfide toxicity, the improved removal of specific toxicants, the enhanced quality of digested sludge or another should be sufficient reason for the application of microaerobic conditions at anaerobic digestion of sludge.

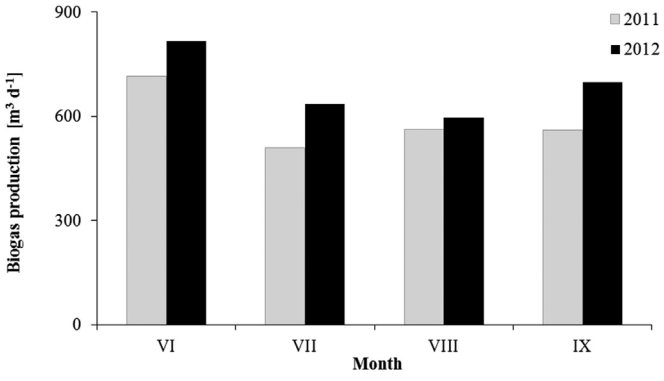



**Fig. 5.** Comparison of the biogas production before (2011) and after (2012) the application of microaeration technology in digester M from June to September.

## 4. Conclusions

The results of long-term full-scale operation of sludge digesters in microaerobic conditions confirm that the biogas desulfurization by microaeration become mature technology and the lack of operating experience has been overcome:

- Main benefits from microaerobic sulfide removal are the ability to remove sulfide inside the anaerobic reactor without requirements to build a new separate desulfurization unit and without additional chemicals.
- The application of the microaerobic conditions is a simple, highly efficient and stable method for hydrogen sulfide removal from the biogas.
- The start-up of the microaerobic biogas desulfurization takes a few weeks due to the necessary air dose optimization and sulfide oxidizing bacteria adaptation.
- The hydrogen sulfide removal efficiency of about 99% can be achieved at a high initial $H_2S$ concentration (above 4000 mg $m^{-3}$). The achievement of hydrogen sulfide concentration in tens of mg $m^{-3}$ is realistic.
- The decrease of the methane content in the biogas due to the presence of surplus nitrogen from the dosed air is lower than expected (maximum 2%). Therefore it is not necessary to use pure oxygen instead of air dosing.
- The VSS/TSS ratio of the digested sludge decreased in most of the cases with the microaerobic conditions due to the better efficiency in VSS degradation.
- A decrease of the soluble COD concentration in the sludge liquor was observed in all systems where the microaerobic conditions were applied.

## Acknowledgement

This research was financially supported by The Czech Ministry of Education, Youth and Sports – project MSM6046137308.

## References

[1] L. Appels, J. Baeyens, J. Degrève, R. Dewil, Principles and potential of the anaerobic digestion of waste-activated sludge, Prog. Energy Combust. Sci. 34 (6) (2008) 755–781.
[2] C. Buisman, R. Post, P. Ijspeert, G. Geraats, G. Lettinga, Biotechnological process for sulphide removal with sulphur reclamation, Acta Biotechnol. 9 (3) (1989) 255–267.
[3] C.A. Celis, Improvement of anaerobic digestion by using of microaerobic conditions, Vol. PhD, in: Department of Water Technology and Environmental Engineering, Institute of Chemical Tcehnology in Prague, Prague, 2012, p. 181.
[4] I. Díaz, A.C. Lopes, S.I. Pérez, M. Fdz-Polanco, Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion, Bioresour. Technol. 101 (20) (2010) 7724–7730.
[5] I. Díaz, S.I. Pérez, E.M. Ferrero, M. Fdz-Polanco, Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters, Bioresour. Technol. 102 (4) (2011) 3768–3775.
[6] P. Jenicek, C.A. Celis, J. Koubova, D. Pokorna, Comparison of microbial activity in anaerobic and microaerobic digesters, Water Sci. Technol. 63 (10) (2011) 2244–2249.
[7] P. Jenicek, C.A. Celis, L. Krayzelova, N. Anferova, D. Pokorna, Improving products of anaerobic sludge digestion by microaeration, Water Sci. Technol. 69 (4) (2014) 803–809.
[8] P. Jenicek, J. Horejs, Full scale experience with biological biogas desulfurization, in: Proc. Of Conf. Solution of Extreme Requirements on WWTP, 2013, pp. 81–88. Blansko, Czech Republic.
[9] P. Jenicek, F. Keclik, J. Maca, J. Bindzar, Use of microaerobic conditions for the improvement of anaerobic digestion of solid wastes, Water Sci. Technol. 58 (7) (2008) 1491–1496.
[10] P. Jenicek, J. Kutil, O. Benes, V. Todt, J. Zabranska, M. Dohanyos, Energy self-

sufficient sewage wastewater treatment plants: is optimized anaerobic sludge digestion the key? Water Sci. Technol. 68 (8) (2013) 1739–1744.
[11] J.E. Johansen, R. Bakke, Enhancing Hydrolysis with Microaeration, vol. 53, 2006, pp. 43–50.
[12] S.K. Khanal, J.C. Huang, Anaerobic treatment of high sulfate wastewater with oxygenation to control sulfide toxicity, J. Environ. Eng. ASCE 129 (12) (2003) 1104–1111.
[13] T. Kobayashi, Y.Y. Li, K. Kubota, H. Harada, T. Maeda, H.Q. Yu, Characterization of sulfide-oxidizing microbial mats developed inside a full-scale anaerobic digester employing biological desulfurization, Appl. Microbiol. Biotechnol. 93 (2) (2012) 847–857.
[14] L. Krayzelova, J. Bartacek, I. Díaz, D. Jeison, E.P. Volcke, P. Jenicek, Microaeration for hydrogen sulfide removal during anaerobic treatment: a review, Rev. Environ. Sci. Bio Technol. 14 (4) (2015) 703–725.
[15] L. Krayzelova, J. Bartacek, N. Kolesarova, P. Jenicek, Microaeration for hydrogen sulfide removal in UASB reactor, Bioresour. Technol. 172 (2014) 297–302.
[16] R. Muñoz, L. Meier, I. Diaz, D. Jeison, A review on the state-of-the-art of physical/chemical and biological technologies for biogas upgrading, Rev. Environ. Sci. Bio Technol. 14 (4) (2015) 727–759.
[17] I. Ramos, I. Diaz, M. Fdz-Polanco, The role of the headspace in hydrogen sulfide removal during microaerobic digestion of sludge, Water Sci. Technol. 66 (10) (2012) 2258–2264.
[18] I. Ramos, R. Perez, M. Fdz-Polanco, The headspace of microaerobic reactors: sulphide-oxidising population and the impact of cleaning on the efficiency of biogas desulphurisation, Bioresour. Technol. 158 (2014) 63–73.
[19] E. Rodriguez, A. Lopes, M. Fdz-Polanco, A.J. Stams, P.A. Garcia-Encina, Molecular analysis of the biomass of a fluidized bed reactor treating synthetic vinasse at anaerobic and micro-aerobic conditions, Appl. Microbiol. Biotechnol. 93 (5) (2012) 2181–2191.
[20] L. Spinosa, A. Ayol, J.-C. Baudez, R. Canziani, P. Jenicek, A. Leonard, W. Rulkens, G. Xu, L. Van Dijk, Sustainable and innovative solutions for sewage sludge management, Water 3 (2) (2011) 702–717.
[21] M. Syed, G. Soreanu, P. Falletta, M. Béland, Removal of hydrogen sulfide from gas streams using biological processes - a review, Can. Biosyst. Eng. 48 (2006), 2.1–2.14.
[22] F.P. van der Zee, S. Villaverde, P.A. Garcia, F. Fdz-Polanco, Sulfide removal by moderate oxygenation of anaerobic sludge environments, Bioresour. Technol. 98 (3) (2007) 518–524.
[23] D.A.J. Wase, C.F. Forster, Biogas — fact or fantasy, Biomass 4 (2) (1984) 127–142.
[24] D.H. Zitomer, J.D. Shrout, High-sulfate, high-chemical oxygen demand wastewater treatment using aerated methanogenic fluidized beds, Water Environ. Res. 72 (1) (2000) 90–97.